

TROITSKO-PECHORSK, Russian Federation

WMO: 237110

Lat: 62.7324N

Lon: 56.1930E

Elev: 136

StdP: 99.70

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.9	-33.6	-39.9	0.1	-36.8	-36.8	0.1	-33.6	6.5	-9.2	5.7	-9.7	0.7	180	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	28.6	19.4	26.3	18.2	24.0	16.9	20.4	26.5	19.2	24.7	18.0	22.5	2.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.3	23.0	17.1	12.4	21.8	16.0	11.6	20.8	59.1	26.9	55.2	25.0	51.3	22.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	5.2	4.5	-38.7	31.2	4.0	1.7	-41.6	32.4	-44.0	33.4	-46.2	34.3	-49.2	35.6		
			-38.3	22.0	4.3	1.2	-41.4	22.9	-43.9	23.6	-46.4	24.3	-49.5	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-16.9	-15.3	-7.1	-0.1	6.5	12.9	16.4	12.7	7.2	0.5	-8.3	-13.6
	DBStd	12.94	8.96	8.62	6.03	5.50	5.30	5.16	4.77	4.23	3.95	4.61	7.76	9.72
	HDD10.0	4238	833	707	528	306	138	28	6	19	101	294	548	730
	HDD18.3	6869	1091	941	787	554	369	177	96	183	333	552	798	988
	CDD10.0	468	0	0	0	2	28	113	204	102	18	1	0	0
	CDD18.3	57	0	0	0	0	1	13	36	8	0	0	0	0
	CDH23.3	603	0	0	0	0	24	146	361	71	1	0	0	0
	CDH26.7	157	0	0	0	0	4	36	99	18	0	0	0	0
Wind	WSAvg	2.3	2.1	2.2	2.6	2.6	2.5	2.4	2.1	2.1	2.2	2.4	2.2	2.2
Precipitation	PrecAvg	633	42	31	29	37	54	60	70	79	61	66	55	47
	PrecMax	816	65	60	74	109	145	164	145	135	109	122	90	89
	PrecMin	517	17	3	7	14	9	15	13	18	18	16	15	14
	PrecStd	71	11	13	16	18	26	30	30	31	23	25	19	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.3	7.2	18.3	27.0	30.1	31.4	29.3	20.4	13.3	6.0	2.3
		MCWB	-0.1	0.4	2.2	10.5	15.7	19.2	20.4	20.0	14.8	10.2	5.6	1.9
	2%	DB	-1.4	-0.2	4.1	12.8	22.8	27.1	29.4	24.9	17.2	10.7	3.1	0.5
		MCWB	-1.9	-0.9	0.7	7.0	13.4	18.1	19.9	18.0	12.6	9.1	2.5	0.0
	5%	DB	-3.0	-1.6	2.3	10.0	19.3	24.5	27.4	22.2	15.2	8.6	1.4	-0.5
		MCWB	-3.6	-2.3	0.1	4.9	11.7	16.8	19.1	16.2	11.9	7.1	1.0	-0.9
	10%	DB	-5.2	-3.5	0.8	7.6	16.3	21.8	25.2	19.8	13.2	6.8	0.5	-2.0
		MCWB	-5.7	-4.1	-0.7	3.5	10.1	15.2	18.1	15.4	10.8	5.8	0.1	-2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.6	3.1	10.7	16.1	21.0	22.2	20.7	15.6	11.1	5.6	1.7
		MCDB	0.1	1.0	5.9	17.2	24.2	27.0	29.0	27.5	19.4	12.5	6.0	2.0
	2%	WB	-1.8	-0.7	1.5	7.3	14.4	19.2	20.9	19.1	13.5	9.0	2.5	0.1
		MCDB	-1.3	-0.2	3.2	12.1	21.4	25.4	27.0	23.9	15.9	10.3	3.0	0.4
	5%	WB	-3.5	-2.3	0.5	5.5	12.5	17.7	19.8	17.3	12.3	7.3	1.1	-0.9
		MCDB	-3.0	-1.7	1.9	8.8	18.1	23.0	25.5	20.5	14.6	8.5	1.4	-0.6
	10%	WB	-5.7	-4.0	-0.5	4.1	10.7	16.1	18.8	15.9	11.0	5.8	0.2	-2.5
		MCDB	-5.2	-3.4	0.9	7.0	15.6	20.8	24.0	18.9	13.0	6.7	0.5	-2.1
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.6	9.0	9.1	10.0	10.0	10.4	8.6	6.8	4.4	5.0	6.0
		MCDBR	7.1	5.0	8.4	12.7	16.0	13.8	13.9	13.1	10.0	6.7	3.8	4.1
		MCWBR	7.0	4.8	6.2	8.1	8.5	6.5	5.8	6.4	5.9	5.0	3.9	3.9
	5% WB	MCDBR	7.0	4.8	6.9	11.2	14.2	12.9	12.4	11.1	8.7	6.4	3.9	4.1
		MCWBR	7.0	4.6	5.4	7.7	8.3	6.5	5.8	6.1	5.6	5.1	4.0	4.1
Clear-Sky Solar Irradiance	taub	0.247	0.294	0.331	0.358	0.360	0.366	0.403	0.381	0.340	0.312	0.249	0.234	
	taud	1.938	2.025	2.027	2.084	2.374	2.439	2.337	2.375	2.458	2.318	2.013	1.911	
	Ebn at Noon	358	616	743	812	856	857	807	793	762	620	357	178	
	Edh at Noon	37	74	109	127	105	100	109	96	73	56	33	19	
All-Sky Solar Radiation	RadAvg	0.23	0.96	2.33	3.95	4.97	5.11	5.19	3.50	1.96	0.73	0.24	0.10	
	RadStd	0.02	0.14	0.26	0.40	0.41	0.37	0.48	0.44	0.26	0.09	0.05	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		+0.98	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page